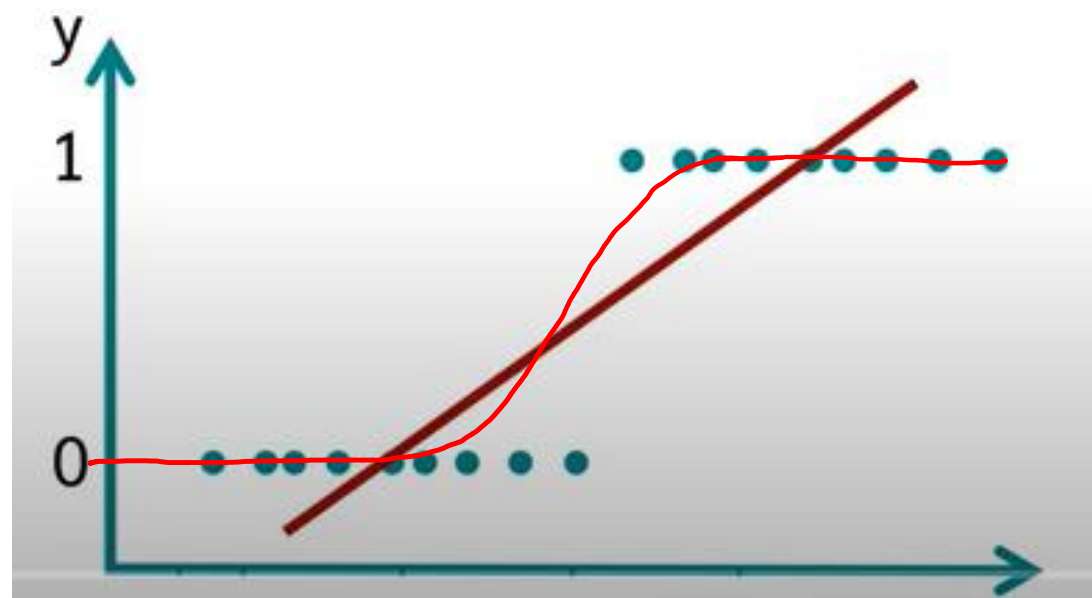
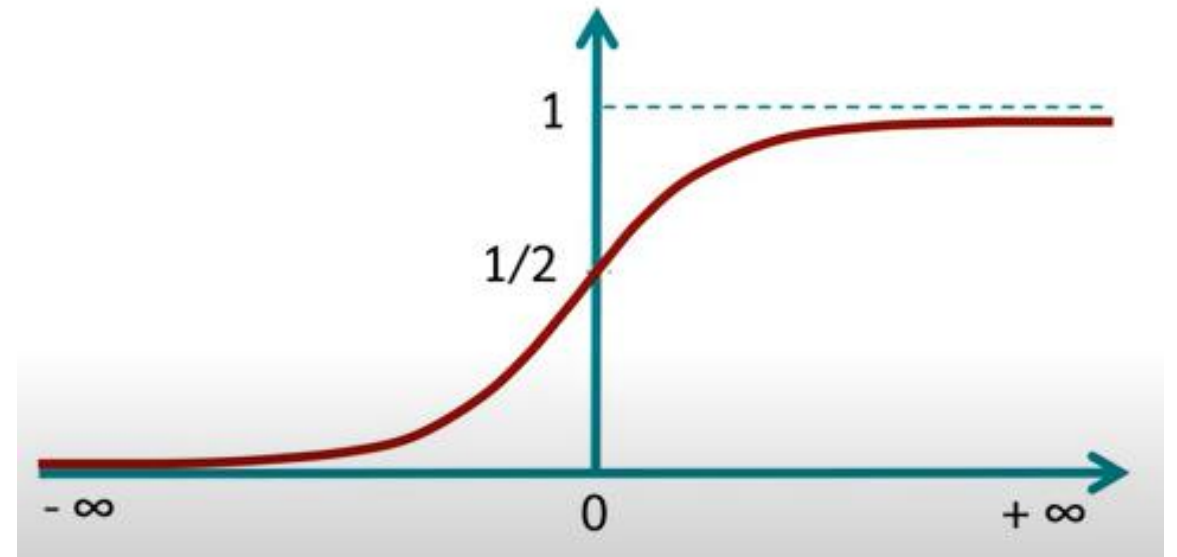


Regresia logistica



Regresia logistica

- Este o metoda de clasificare
- - binara: atribut tinta are valorile 0 sau 1
- Estimeaza care este probabilitatea ca atributul tinta sa aiba valoarea 0 sau 1
- Avem nevoie de o functie care sa ia valori doar in $[0,1]$
- Functia logistica



$$f(z) = \frac{1}{1 + e^{-z}}$$

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$$z = a + b_1 x_1 + b_2 x_2 + \dots + b_n x_n.$$

in 2D.

$$z = a + bx.$$

$$P(y = 1 | x_1, \dots, x_n) = \frac{1}{1 + e^{-(b_1 x_1 + \dots + b_n x_n + a)}}$$

Coeficientii b_i si a se determina prin metoda maximum likelihood method

$$P(y=1/x) = \frac{1}{1 + e^{-(a+bx)}}$$

$$p = p(y=1/x) \Rightarrow 1-p = p(y=0/x)$$

- Se minimizeaza functia cost

$$S = -p \ln p - (1-p) \ln(1-p)$$

entropy loss function
loss function

- Se foloseste metoda gradientului (gradient descent):

gradient.

- Alpha = learning rate = constanta

- $b = b - \alpha * \text{Deriv } S / \text{deriv } b$

- $a = a - \alpha * \text{Deriv } S / \text{deriv } a$

$$\begin{pmatrix} a^{(k+1)} \\ b^{(k+1)} \end{pmatrix} = \begin{pmatrix} a^{(k)} \\ b^{(k)} \end{pmatrix} - \alpha \cdot \begin{bmatrix} \frac{\partial S}{\partial a} \\ \frac{\partial S}{\partial b} \end{bmatrix} \begin{pmatrix} a^{(k)} \\ b^{(k)} \end{pmatrix}$$

Se itereaza pana cand nu se mai modifica a si b